BUSINESS REQUIREMENTS DOCUMENT

E-Commerce Supply Chain Optimisation

ShopFast Analytics Initiative

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1. Executive Summary

ShopFast, a mid-sized e-commerce company processing 50,000+ monthly orders across Brazil, is experiencing a critical deterioration in supply chain performance. Average order fulfillment time has increased from 2.1 days to 3.8 days over the past 6 months — an 81% increase — resulting in an 18% rise in customer complaints and a 12% increase in order cancellations.

The Operations team currently relies on a manual reporting process that takes 3 days to compile, making the data outdated before it reaches management. Without real-time visibility into supply chain bottlenecks, proactive intervention is impossible.

This Business Requirements Document defines the scope, objectives, stakeholders, functional requirements, and success criteria for the Supply Chain Analytics & Optimization initiative. The project will deliver:

- A SQL-based analysis of 100,000+ order records across 9 relational tables to identify the top bottlenecks causing fulfillment delays
- A set of 6 operational KPIs including OTIF rate, fill rate, and average fulfillment time
- An interactive Power BI supply chain dashboard with 4 views for daily operations monitoring
- Three data-backed business recommendations with estimated revenue and efficiency impact

Business case summary

Current avg fulfillment time: 3.8 days | Baseline: 2.1 days | Gap: 1.7 days (81% above target)
Order cancellation rate increase: 12% | Customer complaints increase: 18%
Manual reporting lag: 3 days — data is stale before management sees it
Estimated revenue at risk from cancellations: $42,000+ annually
Target: Reduce avg fulfillment time to ≤3.0 days within 3 months of implementation

2. Project Background & Problem Statement

2.1 Business Context

ShopFast operates across multiple product categories including electronics, fashion, home goods, and health products, serving customers across all Brazilian states. The company works with 3,095 unique sellers and uses multiple shipping partners for last-mile delivery. Growth in order volume over the past 18 months has outpaced the company's supply chain infrastructure, creating systemic bottlenecks that are now directly impacting customer satisfaction and revenue.

2.2 Problem Statement

Core business problem

ShopFast lacks a data-driven visibility system to identify WHERE in the supply chain delays occur.
The current situation creates three compounding problems:

1. Reactive operations: Teams find out about delays after customers complain, not before
2. No seller accountability: Underperforming sellers continue without consequence
3. Inventory blind spots: Stockout-prone categories are not identified until orders fail

Result: $42,000+ annual revenue at risk, rising cancellations, falling customer satisfaction

2.3 Business Opportunity

By implementing a real-time supply chain analytics dashboard powered by SQL analysis of existing order data, ShopFast can reduce average fulfillment time by 0.8 days, improve OTIF rate from 72% to 85%+, and identify and remediate underperforming sellers — protecting an estimated $42,000+ in annual recurring revenue while reducing customer complaint volume by an estimated 20%.

3. Project Objectives & Goals

3.1 Primary Objectives

- **OBJ-01:** Analyze 100,000+ order records across 9 relational database tables using SQL to identify the top 5 supply chain bottlenecks by seller, product category, and geographic region
- **OBJ-02:** Define and calculate 6 operational KPIs — OTIF rate, average fulfillment time, order cancellation rate, fill rate by category, delay rate by seller, and revenue at risk
- **OBJ-03:** Build a 4-view interactive Power BI dashboard enabling daily operations monitoring, replacing the current 3-day manual reporting process
- **OBJ-04:** Deliver 3 business recommendations with projected impact on fulfillment time, OTIF rate, and annual revenue protected
- **OBJ-05:** Document all data quality findings, table relationships, and analytical assumptions in a structured, stakeholder-ready format

3.2 Business Goals

Business goal	Expected outcome
Proactive delay prevention	Identify at-risk orders before customers complain
Seller accountability	Flag bottom 10% sellers for performance improvement
Inventory optimization	Reduce stockout-driven delays in top 3 categories
Reporting efficiency	Replace 3-day manual process with live dashboard
Revenue protection	Protect $42,000+ annual revenue from cancellations

4. Project Scope

4.1 In Scope

- SQL analysis of the Olist Brazilian E-Commerce dataset (100,000+ orders, 9 CSV tables)
- Loading all 9 CSV files into a SQLite database and writing JOIN queries across tables
- Calculation of 6 KPIs using SQL CTEs and window functions
- Python EDA with 4 visualization charts (fulfillment time distribution, delay by category, OTIF trend, seller performance)
- Interactive Power BI dashboard with 4 views published with a shareable link
- 3 business recommendations with projected impact estimates
- GitHub repository with structured folders and professional README

4.2 Out of Scope

- Real-time integration with ShopFast's live order management system
- Predictive delay modelling using machine learning (Phase 2)
- A/B testing of fulfillment strategies
- Primary data collection — analysis limited to the Olist public dataset
- Financial modelling beyond estimated revenue impact figures

4.3 Assumptions

- The Olist Brazilian E-Commerce dataset is an acceptable proxy for ShopFast order data for analytical purposes
- Delivery delay is defined as: actual delivery date > estimated delivery date
- OTIF rate is calculated as: orders delivered on time AND with complete quantity / total orders
- Revenue at risk is estimated at average order value ($140) x cancellation count
- All 9 dataset tables can be joined using the order_id and seller_id keys

4.4 Constraints

- All tools must be free or open-source (Python, SQLite, Power BI Desktop)
- Project must be completable within 3–4 weeks as a solo initiative
- Analysis limited to the static Olist dataset — no live data pipeline

5. Data Sources & Table Dictionary

The Olist Brazilian E-Commerce dataset contains 9 interrelated CSV files. The table below documents each file, its purpose, key fields, and how it connects to other tables.

Table / CSV file	Purpose	Key fields	Joins to
olist_orders	Core order records —	order_id, customer_id, order_status, purchase_timestamp,	All tables via order_id

Table / CSV file	Purpose	Key fields	Joins to
	status, timestamps	delivered_customer_date, estimated_delivery_date	
olist_order_items	Line items per order — seller, product, price	order_id, order_item_id, product_id, seller_id, price, freight_value	orders, sellers, products
olist_sellers	Seller details and location	seller_id, seller_city, seller_state	order_items via seller_id
olist_products	Product catalogue with category	product_id, product_category_name, product_weight_g	order_items via product_id
olist_customers	Customer location data	customer_id, customer_city, customer_state	orders via customer_id
olist_order_payments	Payment method and value	order_id, payment_type, payment_value	orders via order_id
olist_order_reviews	Customer review scores	order_id, review_score, review_comment	orders via order_id
olist_geolocation	Zip code to lat/long mapping	geolocation_zip_code_prefix, lat, lng, city, state	customers, sellers via zip code
product_category_name_translation	Portuguese to English category names	product_category_name, product_category_name_english	products via category name

6. Stakeholder Register

Stakeholder	Role	Interest	Influence	Engagement
Operations Manager	Primary dashboard user	Daily fulfillment visibility	High	Consulted & Informed
Supply Chain Head	Strategy owner	Bottleneck elimination	High	Consulted
Seller Relations Team	Seller performance mgmt	Seller KPI data	Medium	Informed

Stakeholder	Role	Interest	Influence	Engagement
Data Analytics Team	Technical reviewers	SQL accuracy, data quality	Medium	Accountable
Customer Service Team	End beneficiary	Fewer complaint tickets	Low	Informed
Senior Leadership	Budget & strategy approval	Revenue impact, ROI	High	Informed

7. Functional Requirements

7.1 Data & SQL Requirements

ID	Requirement	Acceptance Criteria
DR-01	Load all 9 Olist CSV files into a SQLite database using Python	All 9 tables created; row counts verified against source CSVs
DR-02	Identify and document all data quality issues across tables	Data quality log created; null values, duplicate order_ids, and date format issues resolved
DR-03	Write SQL JOIN queries connecting orders, items, sellers, and products	All 4 core tables successfully joined; query returns correct record counts
DR-04	Calculate fulfillment delay flag for each order	New column: is_late (1 = delivered after estimated date, 0 = on time)
DR-05	Export final analytics dataset as CSV for Power BI	CSV contains all 6 KPI fields; loads cleanly into Power BI without errors

7.2 KPI Requirements

ID	KPI name	SQL formula	Target
KPI-01	OTIF rate	Delivered on time AND complete / total orders	≥ 85%
KPI-02	Avg fulfillment time (days)	AVG(delivered_date - purchase_date)	≤ 3.0 days
KPI-03	Order cancellation rate	Cancelled orders / total orders x 100	≤ 5%
KPI-04	Fill rate by category	Delivered orders / ordered items x 100 per category	≥ 90%
KPI-05	Delay rate by seller	Late orders / total orders per seller x 100	Flag sellers > 25%

ID	KPI name	SQL formula	Target
KPI-06	Revenue at risk	Cancelled orders x avg order value ($140)	Minimize month-on-month

7.3 Dashboard Requirements

ID	Requirement	Acceptance Criteria
DB-01	Supply chain overview: 6 KPI cards showing current vs target values	All 6 KPIs visible; red/amber/green status indicators based on target thresholds
DB-02	Delay analysis by category: bar chart of delay rate per product category	Top 5 delay-prone categories highlighted; sortable by delay rate descending
DB-03	Seller performance table: sortable table of all sellers with OTIF rate and delay rate	Sortable by OTIF rate; bottom 10% sellers highlighted in red
DB-04	Geographic delivery heatmap: map showing avg fulfillment time by Brazilian state	Color-coded by fulfillment time; tooltip shows state name + avg days
DB-05	Dashboard published on Power BI service with shareable URL	Link accessible without login; loads within 8 seconds

8. KPI Dashboard — Current vs Target

KPI	Current baseline	Target	Status indicator rule
OTIF rate	72%	≥ 85%	Green ≥85% | Amber 75–85% | Red <75%
Avg fulfillment time	3.8 days	≤ 3.0 days	Green ≤3.0 | Amber 3.0–3.5 | Red >3.5
Order cancellation rate	12% (rising)	≤ 5%	Green ≤5% | Amber 5–10% | Red >10%
Fill rate by category	78% avg	≥ 90%	Green ≥90% | Amber 80–90% | Red <80%
Delay rate by seller	Top sellers 40%+	Flag >25%	Red flag sellers exceeding 25% delay rate
Revenue at risk (annual)	$42,000+ est.	Decrease MoM	Green = declining | Red = increasing month-on-month

9. Business Recommendations

Based on SQL analysis findings, three strategic recommendations are proposed. These will be validated against dashboard output and presented to the Operations Manager and Supply Chain Head upon project completion.

Recommendation 1 — Seller performance scorecard with automated threshold alerts

Driver addressed: Top 10% worst-performing sellers account for 43% of all late deliveries. Action: Implement a monthly seller scorecard with OTIF rate thresholds. Sellers below 75% OTIF for 2 consecutive months receive a formal warning. Sellers below 60% are flagged for contract review. Projected impact: 15% reduction in seller-caused delays — reducing fulfillment time by an estimated 0.4 days and protecting ~$18,000 annual revenue.

Recommendation 2 — Category-based safety stock buffer for top 3 delay-prone categories

Driver addressed: Top 3 delay-prone product categories (electronics, health & beauty, sports) have stockout rates 2x the average, causing fill rate failures. Action: Increase safety stock by 20% for these 3 categories during peak periods (Q4, promotional events). Trigger automatic reorder when stock falls below 30-day supply. Projected impact: Fill rate improvement from 78% to 88%+ for affected categories — reducing stockout-driven delays by an estimated 25%.

Recommendation 3 — Regional fulfillment partner for high-delay geographic zones

Driver addressed: Northeast and North regions show 40% higher fulfillment times due to single warehouse dependency and longer shipping distances. Action: Partner with a regional fulfillment center in the Northeast to handle orders above 50km from the nearest warehouse. Prioritize orders over $150 for expedited handling. Projected impact: 0.8 day reduction in average fulfillment time for affected regions — bringing company-wide average from 3.8 to ~3.0 days.

10. Project Timeline

Phase	Duration	Key deliverables	Milestone
Phase 1	Week 1 (Days 1–3)	BRD, stakeholder register, KPI definitions, success criteria, scope sign-off	BRD approved
Phase 2	Week 1–2 (Days 4–10)	SQLite database, 9 tables loaded, JOIN queries, KPI calculations, data quality log	SQL analysis complete
Phase 3	Week 2 (Days 11–14)	EDA notebook, 4 Python charts, fulfillment time distribution, seller performance analysis	EDA complete
Phase 4	Week 3 (Days 15–21)	Power BI dashboard (4 views), geographic heatmap, published shareable link	Dashboard live

Phase	Duration	Key deliverables	Milestone
Phase 5	Week 4 (Days 22–28)	Business recommendations report, GitHub repo, README, resume bullets updated	Project complete

11. Risks & Mitigation

Risk	Likelihood	Impact	Mitigation
Missing delivery dates causing incomplete OTIF calculation	High	High	Filter to 'delivered' status only; document excluded orders in data quality log
SQLite performance slow on 100K+ rows	Low	Medium	Use indexed columns for JOINs; limit exploratory queries to sampled datasets first
Power BI geographic map requires Bing Maps — may not render offline	Medium	Low	Test map view with internet connection; use bar chart by state as backup
Revenue impact estimates are approximations	High	Low	Label all figures as estimates; document $140 avg order value assumption clearly
Dataset represents Brazil only — may limit MNC interview relevance	Low	Low	Frame project as methodology demonstration, not country-specific analysis

12. Document Approval & Sign-Off

Name	Role	Signature	Date
Shrusti Khadji	Business Analyst (Author)	_____	May 2025
[Operations Manager]	Primary Stakeholder	_____	
[Supply Chain Head]	Sponsor	_____	
[Data Analytics Lead]	Technical Reviewer	_____	

Appendix A — Version History

Version	Date	Author	Changes
v1.0	May 2025	Shrusti Khadji	Initial release — all sections complete

Appendix B — Glossary

Term	Definition
OTIF	On Time In Full — a supply chain KPI measuring the percentage of orders delivered both on time and with the complete quantity ordered
Fill rate	The percentage of customer orders fulfilled without stockouts or partial shipments
Fulfillment time	The number of days between a customer placing an order and receiving the delivery
SQLite	A lightweight, file-based relational database management system used to store and query structured data using SQL
CTE	Common Table Expression — a temporary named result set in SQL that makes complex queries more readable and reusable
KPI	Key Performance Indicator — a measurable value used to evaluate the success of a business objective
Safety stock	Extra inventory held to prevent stockouts caused by demand variability or supply delays
Last-mile delivery	The final step in the delivery process — from a distribution hub to the customer's address
Power BI	Microsoft's business intelligence tool used to connect data sources, build interactive reports, and publish dashboards
DAX	Data Analysis Expressions — the formula language used in Power BI to create calculated columns and measures